

线

性

代

数

行列式

主对角线

副对角线

对于二元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} \neq 0 \quad (\text{由方程组系数确定的二阶行列式})$$

$$x_1 = \frac{D_1}{D} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{D_2}{D} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

三阶行列式

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

全排列：将 $1, 2, \dots, n$ 这 n 个数任意组合后排成的数组 $j_1 j_2 \dots j_n$

(n 阶全排列，简称“排列”)

$$\text{总数 } P_n = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1 = n!$$

逆序：在全排列中任取两个数，前一个大于后一个，即构成

逆序数：一个全排列中所有逆序的总和 $t = t_1 + t_2 + \dots + t_n = \sum_{k=1}^n t_k$

奇(偶)排列：逆序数为奇(偶)数的全排列

例. 求下列全排列的逆序数

$$(1) \ 312 \quad t = 1 + 1 + 0 = 2$$

$$(2) \ 134782695 \quad t = 4 + 2 + 4 = 10$$

$$(3) \ n(n-1) \dots 321 \quad t_k = t_n - k \ (k=1, 2, \dots, n) \quad t = (n-1) + (n-2) + \dots + 1 + 0 = \frac{(1+n-1)(n-1)}{2} = \frac{n(n-1)}{2}$$

$$(4) \ 123 \dots (n-1)n \quad t_k = 0 \ (k=1, 2, \dots, n) \quad t = 0$$

n 阶行列式

- 共有 $n!$ 项，正负项各为 $\frac{n!}{2}$ 项
- 每项均为取自不同行不同列三个元素的乘积
- 确定符号的法则：当该项元素的行标按自然数顺序（如 $a_{11}a_{22}\dots a_{nn}$ ）排列后，若对应的列标构成偶排列则该项取正号，构成奇排列则取负号

设有 n^2 个数，排成 n 行 n 列的数表

$$\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{array}$$

共有 $n!$ 项，每一项形如 $(-1)^t a_{11}a_{22}\dots a_{nn}$ ★

n 阶行列式 = $n!$ 项代数和 = $\sum (-1)^t a_{11}a_{22}\dots a_{nn}$

记为 $D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = \sum (-1)^t a_{11}a_{22}\dots a_{nn}$, 简记为 $\det(a_{ij})$, a_{ij} 为行列式元素

注：一阶行列式 $|a| = a$, 如 $|-2| = -2$

行列式的几何意义：

(1) $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ 的绝对值表示以向量 $\{a_{11}, a_{12}\}$, $\{a_{21}, a_{22}\}$ 为邻边的平行四边形的面积

(2) $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ 的绝对值表示以向量 $\{a_{11}, a_{12}, a_{13}\}$, $\{a_{21}, a_{22}, a_{23}\}$, $\{a_{31}, a_{32}, a_{33}\}$ 为邻棱的平行六面体的体积

(3) $\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$ 的绝对值表示对应向量组按平行四边形法则组合成的超空间立体的体积

若 n 阶行列式 D 中非零元素的个数小于 n ，则 $D=0$

上三角行列式

下三角行列式

对角行列式

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ & a_{22} & \dots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{vmatrix}$$

$$\left. \begin{array}{l} \text{上三角行列式} \\ \text{下三角行列式} \\ \text{对角行列式} \end{array} \right\} \begin{array}{l} = (-1)^0 a_{11}a_{22}\dots a_{nn} \\ = a_{11}a_{22}\dots a_{nn} \end{array}$$

反对角行列式

$$\begin{vmatrix} & & & a_{1n} \\ & & a_{2(n-1)} & \\ & \ddots & & \\ a_{n1} & & & \end{vmatrix} = (-1)^{\frac{n(n-1)}{2}} a_{1n}a_{2(n-1)}\dots a_{n1}$$

对换：将一个排列中的两个数位置调换，将相邻两个数对换称为相邻对换

对换与排列的奇偶性有直接关系

• 一个排列中的任意两个元素对换，排列改变奇偶性

• 标准排列 (如 $123\dots n$) 是偶排列，逆序数是 0

• 奇排列变成标准排列的对换次数为奇数，偶排列变成标准排列的对换次数为偶数

• 在所有 n 阶排列中，奇、偶排列各为 $\frac{n!}{2}$ 个

• n 阶行列式亦可定义为 $D = \sum (-1)^S a_{1j_1} a_{2j_2} \dots a_{nj_n}$

$S = t(i_1 i_2 \dots i_n) + t(j_1 j_2 \dots j_n)$ ，为行标与列标排列的逆序数之和

两种方法：

例. 在六阶行列式中，下列两项各应带什么号？

(1) $a_{23} a_{31} a_{42} a_{56} a_{14} a_{65}$

(2) $a_{32} a_{43} a_{14} a_{51} a_{66} a_{25}$

原式 = $a_{14} a_{23} a_{31} a_{42} a_{56} a_{65}$

431265 的逆序数 $t = 1+2+2+1 = 6$ 为偶数

带正号

行标 341562 的逆序数为 $t_1 = 4+2 = 6$

列标 234165 的逆序数为 $t_2 = 1+3 = 4$

$S = t_1 + t_2 = 10$ 为偶数 带正号

例. 用行列式定义计算

$$D_n = \begin{vmatrix} 0 & 0 & \dots & 0 & 1 & 0 \\ 0 & 0 & \dots & 2 & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ n-1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 & n \end{vmatrix}$$

∵ 其中不为零的不同行不同列的乘积只有 $a_{1,(n-1)} a_{2,(n-2)} \dots a_{n-1,1} a_{nn} = n!$

而 $t[(n-1)(n-2)\dots 1 \cdot n] = (n-2) + (n-3) + \dots + 1 + 0 = \frac{(1+n-2)(n-2)}{2} = \frac{(n-1)(n-2)}{2}$

∴ $D_n = (-1)^{\frac{(n-1)(n-2)}{2}} n!$

小练习 Number 1

(1) 求下列排列的逆序数：

$$13 \cdots (2n-1)24 \cdots (2n)$$

$$13 \cdots (2n-1)(2n)(2n-2) \cdots 2$$

排列 1 3 ... (2n-1) 2 4 ... (2n-2) 2n

排列 1 3 ... (2n-1) (2n) (2n-2) ... 4 2

逆序 0 0 ... 0 n-1 n-2 ... 1 0

逆序 0 0 ... 0 0 2 ... 2n-4 2n-2

$$t = (n-1) + (n-2) + \cdots + 1 = \frac{(1+n-1)(n-1)}{2} = \frac{n(n-1)}{2}$$

$$t = 2 + 4 + \cdots + (2n-4) + (2n-2) = \frac{(2+2n-2)(n-1)}{2} = n(n-1)$$

(2) 写出四阶行列式中含有因子 $a_{11}a_{23}$ 的项

$$a_{11}a_{23}a_{34}a_{42}$$

$$- a_{11}a_{23}a_{32}a_{44}$$

(3) 选择 k, l , 使 $a_{13}a_{2k}a_{34}a_{42}a_{5l}$ 成为五阶行列式 $|a_{ij}|$ 中带负号的项

$$k=1, l=5$$

(4) 设 n 阶行列式中有 n^2-n 个以上的元素为零, 证明该行列式为 0.

证明: ① 已知 n 阶行列式每行恰有 n 个元素, 共有 n^2 个元素

若有 n^2-n 个以上元素为 0, 则必满足至少有一行元素都是 0

由行列式性质得其值为 0.

或 ② 由题意知 n 阶行列式 D 中非零元素的个数少于 n ,

$$D \text{ 的一般项 } a_{1j_1}a_{2j_2}\cdots a_{nj_n} = 0$$

由行列式定义得 $D=0$

(5) 用行列式的定义计算下列行列式

$$\begin{vmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{vmatrix}$$

$$\text{设原式} = |a_{ij}| \quad (i, j = 1, 2, 3, 4)$$

$|a_{ij}|$ 的展开式中除 $a_{13}a_{22}a_{34}a_{41}$ 这一项外,

其余各项都含零元素, 结果皆为零

$$\text{故 } |a_{ij}| = (-1)^{t(3241)} a_{13}a_{22}a_{34}a_{41} = 1$$

$$\begin{vmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 \\ n & 0 & 0 & \cdots & 0 \end{vmatrix}$$

$$\text{设原式} = |a_{ij}| \quad (i, j = 1, 2, \dots, n)$$

$|a_{ij}|$ 展开式中仅 $a_{12}a_{23}\cdots a_{n-1,n}a_{n1}$

这一项不为 0

$$\text{故 } |a_{ij}| = (-1)^{t(23\cdots n1)} a_{12}a_{23}\cdots a_{n-1,n}a_{n1} = (-1)^{n-1} n!$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & 0 & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 & 0 \\ a_{51} & a_{52} & 0 & 0 & 0 \end{vmatrix}$$

$$\text{设原式} = |a_{ij}| \quad (i, j = 1, 2, \dots, 5)$$

$|a_{ij}|$ 第 5 行仅 $a_{51}, a_{52} \neq 0$

第 4 行仅 $a_{41}, a_{42} \neq 0$

第 3 行仅 $a_{31}, a_{32} \neq 0$

不论取 $a_{51}a_{42}$ 还是 $a_{52}a_{41}$

第 3 行元素均不能取非零的第

1 列、第 2 列, 但 $a_{33}, a_{34}, a_{35} = 0$

故 $|a_{ij}|$ 各项都为 0

$$\therefore |a_{ij}| = 0$$

行列式的性质

$$D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} \xrightarrow{\text{转置行列式}} D^T = \begin{vmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{vmatrix}$$

$$= \det(a_{ij}) \qquad \qquad \qquad = \det(b_{ij})$$

$$a_{ij} = b_{ji} \qquad b_{ij} = a_{ji} \qquad (i, j = 1, 2, \dots, n)$$

• $D^T = \sum (-1)^t b_{1j_1} b_{2j_2} \dots b_{nj_n} = \sum (-1)^t a_{j_1 1} a_{j_2 2} \dots a_{j_n n} = D$

• 交换两行(列), 行列式仅改变符号

$$D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} \qquad D_1 = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

$$S = t(1 \dots i \dots j \dots n) + t(p_1 \dots p_i \dots p_j \dots p_n)$$

$$S_1 = t(1 \dots j \dots i \dots n) + t(p_1 \dots p_j \dots p_i \dots p_n)$$

S 与 S_1 奇偶性相反

$$D = \sum (-1)^S a_{1p_1} \dots a_{ip_i} \dots a_{jp_j} \dots a_{np_n}$$

$$D_1 = \sum (-1)^{S_1} a_{1p_1} \dots a_{jp_j} \dots a_{ip_i} \dots a_{np_n}$$

$$\therefore D = -D_1 \qquad D_1 = -D$$

一般用 r_i 表示第 i 行, 用 c_j 表示第 j 列, 交换行记作 $r_i \leftrightarrow r_j$, 交换列记作 $c_i \leftrightarrow c_j$

• 若行列中有两行(列)完全相同, 此行列式 $D = 0$

• 用数 k 乘行列式的某一行(列), 等于用数 k 乘此行列式, 即

$$D_1 = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ ka_{i1} & ka_{i2} & \dots & ka_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = kD$$

用数 k 乘第 i 行(列)通常记作 $r_i \times k$ (或 $c_i \times k$)

例. 设 $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 1$, 求 $\begin{vmatrix} 6a_{11} & -2a_{12} & -10a_{13} \\ -3a_{21} & a_{22} & 5a_{23} \\ -3a_{31} & a_{32} & 5a_{33} \end{vmatrix}$

解: 原式 = $-2 \begin{vmatrix} -3a_{11} & a_{12} & 5a_{13} \\ -3a_{21} & a_{22} & 5a_{23} \\ -3a_{31} & a_{32} & 5a_{33} \end{vmatrix} = (-2) \times (-3) \times 5 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 30$

反对称行列式

形如:

$$D = \begin{vmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ -a_{12} & 0 & a_{23} & \cdots & a_{2n} \\ -a_{13} & -a_{23} & 0 & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_{1n} & -a_{2n} & -a_{3n} & \cdots & 0 \end{vmatrix}$$

- 奇数阶反对称行列式的值为0

$$\therefore D = D^T = \begin{vmatrix} 0 & -a_{12} & -a_{13} & \cdots & -a_{1n} \\ a_{12} & 0 & -a_{23} & \cdots & -a_{2n} \\ a_{13} & a_{23} & 0 & \cdots & -a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & 0 \end{vmatrix} = (-1)^n \begin{vmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ -a_{12} & 0 & a_{23} & \cdots & a_{2n} \\ -a_{13} & -a_{23} & 0 & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_{1n} & -a_{2n} & -a_{3n} & \cdots & 0 \end{vmatrix} = (-1)^n D = -D$$

$$\therefore D = 0$$

- 行列式中若有两行(列)对应元素成比例, 则此行列式为0

- 若行列式的某一行(列)的元素都是两数之和

例如 $D = \begin{vmatrix} a_{11} & a_{12} & \cdots & (a_{1i} + a_{1i}') & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & (a_{2i} + a_{2i}') & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{ni} & a_{nz} & \cdots & (a_{ni} + a_{ni}') & \cdots & a_{nn} \end{vmatrix}$

则 $D = D_1 + D_2 = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1i} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2i} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{ni} & a_{nz} & \cdots & a_{ni} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1i}' & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2i}' & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{ni} & a_{nz} & \cdots & a_{ni}' & \cdots & a_{nn} \end{vmatrix}$

- 把行列式的某一行(列)的各元素乘以同一个数后加到另一行(列)对应元素上, 行列式不变

例如 $D = \begin{vmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2i} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{ni} & \cdots & a_{ni} & \cdots & a_{nj} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & \cdots & a_{1i} + ka_{1j} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2i} + ka_{2j} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{ni} & \cdots & a_{ni} + ka_{nj} & \cdots & a_{nj} & \cdots & a_{nn} \end{vmatrix} = D_1$

用数k乘第j行(列)加到第i行(列)通常记作 $r_i + kr_j$ ($c_i + kc_j$)

例. 证明: $\begin{vmatrix} b+c & c+a & a+b \\ b+c_1 & c_1+a_1 & a_1+b_1 \\ b_2+c_2 & c_2+a_2 & a_2+b_2 \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$

证明: 左边 = $\begin{vmatrix} b & c+a & a+b \\ b_1 & c_1+a_1 & a_1+b_1 \\ b_2 & c_2+a_2 & a_2+b_2 \end{vmatrix} + \begin{vmatrix} c & c+a & a+b \\ c_1 & c_1+a_1 & a_1+b_1 \\ c_2 & c_2+a_2 & a_2+b_2 \end{vmatrix} = \begin{vmatrix} b & c+a & a \\ b_1 & c_1+a_1 & a_1 \\ b_2 & c_2+a_1 & a_2 \end{vmatrix} + \begin{vmatrix} c & a & a+b \\ c_1 & a_1 & a_1+b_1 \\ c_2 & a_2 & a_2+b_2 \end{vmatrix}$

$$= \begin{vmatrix} b & c & a \\ b_1 & c_1 & a_1 \\ b_2 & c_2 & a_2 \end{vmatrix} + \begin{vmatrix} c & a & b \\ c_1 & a_1 & b_1 \\ c_2 & a_2 & b_2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} + \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

例. 计算行列式

$$\textcircled{1} D = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 1 & 3 & -4 \\ 2 & 0 & 1 & -1 \\ 1 & -5 & 3 & -3 \end{vmatrix}$$

解: $D \xrightarrow{C_1 \leftrightarrow C_2} \begin{vmatrix} 1 & 3 & -1 & 2 \\ 1 & -5 & 3 & -4 \\ 0 & 2 & 1 & -1 \\ -5 & 1 & 3 & -3 \end{vmatrix} \xrightarrow{r_2 - r_1} \begin{vmatrix} 1 & 3 & -1 & 2 \\ 0 & -8 & 4 & -6 \\ 0 & 2 & 1 & -1 \\ -5 & 1 & 3 & -3 \end{vmatrix} \xrightarrow{r_4 + 5r_1} \begin{vmatrix} 1 & 3 & -1 & 2 \\ 0 & -8 & 4 & -6 \\ 0 & 2 & 1 & -1 \\ 0 & 16 & -2 & 7 \end{vmatrix}$

$\xrightarrow{\substack{r_3 + \frac{r_2}{4} \\ r_4 + 2r_2}} \begin{vmatrix} 1 & 3 & -1 & 2 \\ 0 & -8 & 4 & -6 \\ 0 & 0 & 2 & -\frac{5}{2} \\ 0 & 0 & 6 & -5 \end{vmatrix} \xrightarrow{r_4 - 3r_3} \begin{vmatrix} 1 & 3 & -1 & 2 \\ 0 & -8 & 4 & -6 \\ 0 & 0 & 2 & -\frac{5}{2} \\ 0 & 0 & 0 & \frac{5}{2} \end{vmatrix} = -1 \times (-8) \times 2 \times \frac{5}{2} = 40$

$$\textcircled{2} D = \begin{vmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix}$$

解: $D \xrightarrow{r_1 + (r_2 + r_3 + r_4)} \begin{vmatrix} 6 & 6 & 6 & 6 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix} = 6 \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix} \xrightarrow{\substack{r_2 - r_1 \\ r_3 - r_1 \\ r_4 - r_1}} 6 \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} = 6 \times 1 \times 2 \times 2 \times 2 = 48$

$\Rightarrow$ 推广: $\begin{vmatrix} a & b & \cdots & b \\ b & a & \cdots & b \\ \vdots & \vdots & \ddots & \vdots \\ b & b & \cdots & a \end{vmatrix} = [a + (n-1)b](a-b)^{n-1}$

$$\textcircled{3} D = \begin{vmatrix} a_1 & -a_1 & 0 & 0 \\ 0 & a_2 & -a_2 & 0 \\ 0 & 0 & a_3 & -a_3 \\ 1 & 1 & 1 & 1 \end{vmatrix} \xrightarrow{C_2 + C_1} \begin{vmatrix} a_1 & 0 & 0 & 0 \\ 0 & a_2 & -a_2 & 0 \\ 0 & 0 & a_3 & -a_3 \\ 1 & 2 & 1 & 1 \end{vmatrix} \xrightarrow{C_3 + C_2} \begin{vmatrix} a_1 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 \\ 0 & 0 & a_3 & -a_3 \\ 1 & 2 & 3 & 1 \end{vmatrix}$$

$\xrightarrow{C_4 + C_3} \begin{vmatrix} a_1 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 \\ 0 & 0 & a_3 & 0 \\ 1 & 2 & 3 & 4 \end{vmatrix} = 4a_1a_2a_3$

$$\bullet \text{ 设 } D = \begin{vmatrix} a_{11} & \cdots & a_{1k} & & \\ \vdots & & \vdots & & \\ a_{k1} & \cdots & a_{kk} & & \\ c_{11} & \cdots & c_{1k} & b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots & \vdots & & \vdots \\ c_{n1} & \cdots & c_{nk} & b_{n1} & \cdots & b_{nn} \end{vmatrix}, \quad D_1 = \det(a_{ij}) = \begin{vmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{vmatrix}$$

$$D_2 = \det(b_{ij}) = \begin{vmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{k1} & \cdots & b_{kn} \end{vmatrix}$$

$$D = D_1 D_2$$

• 计算 $2n$ 阶行列式

$$D_{2n} = \begin{vmatrix} a & & & & b \\ & \ddots & & & \\ & & a & b & \\ & & c & d & \\ & & & \ddots & \\ c & & & & d \end{vmatrix}$$

把 D_{2n} 中的第 $2n$ 行依次与 $(2n-1)$ 行、 $\dots$ 、 2 行对调 ($2n-2$ 次调换), 得:

$$D_{2n} = \begin{vmatrix} a & & & & b \\ c & a & & b & \\ & \ddots & & & \\ & & a & b & \\ & & c & d & \\ c & & & & d \end{vmatrix} \cdot (-1)^{2n-2}$$

再把第 $2n$ 列依次与第 $2n-1$ 列, $\dots$ 第 2 列对调, 得:

$$D_{2n} = \begin{vmatrix} a & b \\ c & d \\ & a & & b \\ & & \ddots & \\ & & c & d \\ c & & & & d \end{vmatrix} \cdot (-1)^{2n-2} \cdot (-1)^{2n-2} = (-1)^{2(2n-2)}$$

$$D_{2n} = D_2 \cdot D_{2n-2} = (ad-bc) D_{2(n-1)}$$

$\dots$ 依次类推, 得:

$$D_{2n} = (ad-bc)^2 D_{2(n-2)} = \dots = (ad-bc)^{n-1} D_2 = (ad-bc)^n$$

小练习

Number 2

(1) 用行列式的性质计算下列各式

$$\begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix} = adf \begin{vmatrix} -b & c & e \\ b & -c & e \\ b & c & -e \end{vmatrix} = abcdef \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$\begin{matrix} C_2+C_1 \\ C_3+C_1 \end{matrix} abcdef \begin{vmatrix} -1 & 0 & 0 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{vmatrix} = (-1)^1 \times (-4) abcdef = 4 abcdef$$

$a_{11}a_{23}a_{32}$ 奇

$$\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}$$

$$\begin{matrix} r_4-r_3 \\ r_3-r_2 \\ r_2-r_1 \end{matrix} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & -3 & 1 & 1 \end{vmatrix}$$

$$\begin{matrix} r_3-r_2 \\ r_2-r_1 \end{matrix} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & -3 \\ 1 & 1 & -3 & 1 \\ 1 & -3 & 1 & 1 \end{vmatrix}$$

$$\begin{matrix} r_4-r_3 \\ r_3-r_2 \\ r_2-r_1 \end{matrix} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & -3 \\ 1 & 1 & -3 & 1 \\ 0 & -4 & 4 & 0 \end{vmatrix}$$

$$\begin{matrix} r_3-r_2 \\ r_2-r_1 \end{matrix} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -7 \\ 0 & 0 & -4 & 4 \\ 0 & -4 & 4 & 0 \end{vmatrix}$$

$$\text{按 } C_1 \text{ 展开 } 1 \cdot (-1)^{1+1} \begin{vmatrix} -1 & -2 & -7 \\ 0 & -4 & 4 \\ -4 & 4 & 0 \end{vmatrix}$$

$$= 4 \times 4 \begin{vmatrix} -1 & -2 & -7 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{vmatrix}$$

$$= -16 \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} + 32 \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} - 112 \begin{vmatrix} 0 & -1 \\ -1 & 1 \end{vmatrix} = 160$$

$$\begin{vmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix}$$

$$\begin{matrix} r_1+ar_2 \\ r_3+dr_4 \end{matrix} \begin{vmatrix} 0 & 1+ab & a & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix}$$

$$\text{按 } C_1 \text{ 展开 } (-1)(-1)^{2+1} \begin{vmatrix} 1+ab & a & 0 \\ -1 & c & 1 \\ 0 & -1 & d \end{vmatrix}$$

$$\begin{matrix} C_3+dc_2 \\ C_4+dc_3 \end{matrix} \begin{vmatrix} 1+ab & a & ad \\ -1 & c & 1+cd \\ 0 & -1 & 0 \end{vmatrix}$$

$$\text{按 } r_2 \text{ 展开 } (-1)(-1)^{3+2} \begin{vmatrix} 1+ab & ad \\ -1 & 1+cd \end{vmatrix} = (1+ab)(1+cd) + ad$$

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix}$$

$$\begin{matrix} r_2+r_1 \\ r_3+r_1 \\ r_4+r_1 \end{matrix} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{vmatrix}$$

$$\begin{matrix} r_3+r_1 \\ r_4+r_1 \end{matrix} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{vmatrix}$$

$$= 1 \times 2 \times 2 \times 2 = 8$$

(2) 把下列行列式化为上三角行列式, 并计算值

$$\begin{vmatrix} -2 & 2 & -4 & 0 \\ 4 & -1 & 3 & 5 \\ 3 & 1 & -2 & -3 \\ 2 & 0 & 5 & 1 \end{vmatrix} = 2 \begin{vmatrix} -1 & 1 & -2 & 0 \\ 4 & -1 & 3 & 5 \\ 3 & 1 & -2 & -3 \\ 2 & 0 & 5 & 1 \end{vmatrix}$$

$$\begin{array}{l} r_2 + 4r_1 \\ r_3 + 3r_1 \\ r_4 + 2r_1 \end{array} \quad 2 \begin{vmatrix} -1 & 1 & -2 & 0 \\ 0 & 3 & -5 & 5 \\ 0 & 4 & -8 & -3 \\ 0 & 2 & 1 & 1 \end{vmatrix}$$

$$\begin{array}{l} r_3 - 2r_4 \\ r_4 - \frac{2}{3}r_2 \end{array} \quad 2 \begin{vmatrix} -1 & 1 & -2 & 0 \\ 0 & 3 & -5 & 5 \\ 0 & 0 & -10 & -5 \\ 0 & 2 & 1 & 1 \end{vmatrix}$$

$$\begin{array}{l} r_4 - \frac{2}{3}r_2 \\ r_4 + \frac{13}{30}r_3 \end{array} \quad 2 \begin{vmatrix} -1 & 1 & -2 & 0 \\ 0 & 3 & -5 & 5 \\ 0 & 0 & -10 & -5 \\ 0 & 0 & \frac{13}{3} & -\frac{7}{3} \end{vmatrix}$$

$$\begin{array}{l} r_4 + \frac{13}{30}r_3 \\ r_4 + \frac{13}{30}r_3 \end{array} \quad 2 \begin{vmatrix} -1 & 1 & -2 & 0 \\ 0 & 3 & -5 & 5 \\ 0 & 0 & -10 & -5 \\ 0 & 0 & 0 & -\frac{27}{6} \end{vmatrix}$$

$$= 2 \times (-1) \times 3 \times (-10) \times \left(-\frac{27}{6}\right)$$

$$= -270$$

$$\begin{vmatrix} 0 & 4 & 5 & -1 & 2 \\ -5 & 0 & 2 & 0 & 1 \\ 7 & 2 & 0 & 3 & -4 \\ -3 & 1 & -1 & -5 & 0 \\ 2 & -3 & 0 & 1 & 3 \end{vmatrix}$$

$$\begin{array}{l} r_2 + 2r_4 \\ r_1 + 5r_4 \end{array} \quad \begin{vmatrix} -15 & 9 & 0 & -26 & 2 \\ -11 & 2 & 0 & -10 & 1 \\ 7 & 2 & 0 & 3 & -4 \\ -3 & 1 & -1 & -5 & 0 \\ 2 & -3 & 0 & 1 & 3 \end{vmatrix}$$

$$\begin{array}{l} \text{按 } C_3 \text{ 展开} \\ (-1) \times (-1)^{4+3} \end{array} \quad \begin{vmatrix} -15 & 9 & -26 & 2 \\ -11 & 2 & -10 & 1 \\ 7 & 2 & 3 & -4 \\ 2 & -3 & 1 & 3 \end{vmatrix}$$

$$\begin{array}{l} r_1 - 2r_2 \\ r_3 + 4r_2 \\ r_4 - 3r_2 \end{array} \quad \begin{vmatrix} 7 & 5 & -6 & 0 \\ -11 & 2 & -10 & 1 \\ -37 & 10 & -37 & 0 \\ 35 & -9 & 31 & 0 \end{vmatrix}$$

$$\begin{array}{l} r_1 - 2r_2 \\ r_3 + 4r_2 \\ r_4 - 3r_2 \end{array} \quad \begin{vmatrix} 7 & 5 & -6 \\ -37 & 10 & -37 \\ 35 & -9 & 31 \end{vmatrix}$$

$$\begin{array}{l} C_1 - C_3 \\ C_1 - C_3 \end{array} \quad \begin{vmatrix} 13 & 5 & -6 \\ 0 & 10 & -37 \\ 4 & -9 & 31 \end{vmatrix}$$

$$= 13 \begin{vmatrix} 10 & -37 \\ -9 & 31 \end{vmatrix} - 5 \begin{vmatrix} 0 & -37 \\ 4 & 31 \end{vmatrix} - 6 \begin{vmatrix} 0 & 10 \\ 4 & -9 \end{vmatrix}$$

$$= -799$$

(3) 设行列式 $|a_{ij}| = m$ ($i, j = 1, 2, \dots, 5$), 依下列次序对 $|a_{ij}|$ 变换后, 求其结果:

交换 r_1 与 r_5 , 再转置, 用 2 乘所有元素, 再用 $(-3) \times C_2$ 加到 C_4 , 最后用 4 除 r_2 各元素

$$-m$$

$$-m$$

$$-m \times 2^5$$

$$-32m$$

$$-\frac{32m}{4}$$

$$-8m.$$

(4) 用行列式的性质证明下列等式

$$\textcircled{1} \begin{vmatrix} a_1+kb_1 & b_1+c_1 & c_1 \\ a_2+kb_2 & b_2+c_2 & c_2 \\ a_3+kb_3 & b_3+c_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

证明: 左边 $\xrightarrow{C_2-C_3} \begin{vmatrix} a_1+kb_1 & b_1 & c_1 \\ a_2+kb_2 & b_2 & c_2 \\ a_3+kb_3 & b_3 & c_3 \end{vmatrix} \xrightarrow{C_1-kC_2} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$\textcircled{2} \begin{vmatrix} y+z & z+x & x+y \\ x+y & y+z & z+x \\ z+x & x+y & y+z \end{vmatrix} = 2 \begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix}$$

证明: 左边 $\xrightarrow{r_1-(r_2+r_3)} \begin{vmatrix} -2x & -2y & -2z \\ x+y & y+z & z+x \\ z+x & x+y & y+z \end{vmatrix} = -2 \begin{vmatrix} x & y & z \\ x+y & y+z & z+x \\ z+x & x+y & y+z \end{vmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-r_1}} -2 \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} = 2 \begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix}$

$$\textcircled{3} \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} = 0$$

证明: 左边 $\xrightarrow{\substack{C_2-C_1 \\ C_3-C_1 \\ C_4-C_1}} \begin{vmatrix} a^2 & 2a+1 & 4a+4 & 6a+9 \\ b^2 & 2b+1 & 4b+4 & 6b+9 \\ c^2 & 2c+1 & 4c+4 & 6c+9 \\ d^2 & 2d+1 & 4d+4 & 6d+9 \end{vmatrix} \xrightarrow{\substack{C_3-2C_2 \\ C_4-C_3-C_2}} \begin{vmatrix} a^2 & 2a+1 & 2 & 4 \\ b^2 & 2b+1 & 2 & 4 \\ c^2 & 2c+1 & 2 & 4 \\ d^2 & 2d+1 & 2 & 4 \end{vmatrix} = 0$

C_3 与 C_4 对应成比例

(5) 计算下列行列式

$$\textcircled{1} \begin{vmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ -1 & 0 & 3 & \dots & n-1 & n \\ -1 & -2 & 0 & \dots & n-1 & n \\ \dots & \dots & \dots & \dots & \dots & \dots \\ -1 & -2 & -3 & \dots & 0 & n \\ -1 & -2 & -3 & \dots & -(n-1) & 0 \end{vmatrix}$$

$\xrightarrow{\substack{r_2+r_1 \\ r_3+r_1 \\ \dots \\ r_n+r_1}} \begin{vmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 0 & 2 & 6 & \dots & 2(n-1) & 2n \\ 0 & 0 & 3 & \dots & 2(n-1) & 2n \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & n-1 & 2n \\ 0 & 0 & 0 & \dots & 0 & n \end{vmatrix} = n!$

$$\textcircled{2} \begin{vmatrix} 1 & a_1 & a_2 & \dots & a_n \\ 1 & a_1+b_1 & a_2 & \dots & a_n \\ 1 & a_1 & a_2+b_2 & \dots & a_n \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & a_1 & a_2 & \dots & a_n+b_n \end{vmatrix}$$

$\xrightarrow{\substack{r_2-r_1 \\ r_3-r_1 \\ \vdots \\ r_n-r_1}} \begin{vmatrix} 1 & a_1 & a_2 & \dots & a_n \\ 0 & b_1 & 0 & \dots & 0 \\ 0 & 0 & b_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & b_n \end{vmatrix} = b_1 b_2 \dots b_n$

⑤

$$D = \begin{vmatrix} x & a_1 & a_2 & \cdots & a_{n-1} & 1 \\ a_1 & x & a_2 & \cdots & a_{n-1} & 1 \\ a_1 & a_2 & x & \cdots & a_{n-1} & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_1 & a_2 & a_3 & \cdots & x & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n & 1 \end{vmatrix} \begin{array}{l} \\ \\ \hline C_1 - a_1 C_{n+1} \\ C_2 - a_2 C_{n+1} \\ \cdots \\ C_n - a_n C_{n+1} \end{array} \begin{vmatrix} x-a_1 & a_1-a_2 & a_2-a_3 & \cdots & a_{n-1}-a_n & 1 \\ 0 & x-a_2 & a_2-a_3 & \cdots & a_{n-1}-a_n & 1 \\ 0 & 0 & x-a_3 & \cdots & a_{n-1}-a_n & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x-a_n & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$$

$$= (x-a_1)(x-a_2)\cdots(x-a_n)$$

(6) 解下列方程

① $\begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 1 & 9-x^2 \end{vmatrix} = 0$

解:

$$\begin{array}{l} \text{原式} \xrightarrow{C_2-C_1} \begin{vmatrix} 1 & 0 & 2 & 3 \\ 1 & 1-x^2 & 2 & 3 \\ 2 & 1 & 1 & 5 \\ 2 & 1 & 1 & 9-x^2 \end{vmatrix} \\ \xrightarrow{C_3-2C_1} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1-x^2 & 0 & 0 \\ 2 & 1 & -3 & -1 \\ 2 & 1 & -3 & 3-x^2 \end{vmatrix} \\ \xrightarrow{C_4-3C_1} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1-x^2 & 0 & 0 \\ 2 & 1 & -3 & -1 \\ 2 & 1 & -3 & 3-x^2 \end{vmatrix} \\ \xrightarrow{C_4-\frac{1}{3}C_3} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1-x^2 & 0 & 0 \\ 2 & 1 & -3 & 0 \\ 2 & 1 & -3 & 4-x^2 \end{vmatrix} \end{array}$$

$$= |x(1-x^2) \times (-3) \times (4-x^2)| = 0$$

解得 $x_{1,2} = \pm 1, x_{3,4} = \pm 2$

② $\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 1-x & 1 & \cdots & 1 & 1 \\ 1 & 1 & 2-x & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & (n-2)-x & 1 \\ 1 & 1 & 1 & \cdots & 1 & (n-1)-x \end{vmatrix} = 0$

解:

写法: $\leftarrow$ 原式 $\xrightarrow{C_2-C_1} \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1-x & 0 & \cdots & 0 & 0 \\ 1 & 0 & 1-x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & (n-3)-x & 0 \\ 1 & 0 & 0 & \cdots & 0 & (n-2)-x \end{vmatrix}$

$$= -x(1-x)\cdots[(n-3)-x][(n-2)-x] = 0$$

解得 $x_1 = 0, x_2 = 1, x_3 = 2, \dots, x_{n-2} = n-3, x_{n-1} = n-2$

行列式按行/列展开

在 n 阶行列式中, 将 a_{ij} 所在的第 i 行, 第 j 列划去后, 余下的 $(n-1)$ 阶行列式称为 a_{ij} 的余子式, 记为 M_{ij} , 而 $A_{ij} = (-1)^{i+j} M_{ij}$, 称为 a_{ij} 的代数余子式

$$\text{对 } D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$a_{11}, a_{12}, a_{13} \text{ 的余子式分别为 } M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$a_{11}, a_{12}, a_{13} \text{ 的代数余子式分别为 } A_{11} = (-1)^{1+1} M_{11}, A_{12} = (-1)^{1+2} M_{12}, A_{13} = (-1)^{1+3} M_{13}$$

$$D = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

- 一个 n 阶行列式 D , 如果其中第 i 行除 a_{ij} 外所有元素都为零, 则该行列式等于 $a_{ij} \times A_{ij}$, 即 $D = a_{ij}A_{ij}$

- 行列式按行/列展开法则:

行列式 = 某一行/列的各元素与其对应的代数余子式的乘积之和

$$\text{即 } \begin{cases} D = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in} \quad (i=1, 2, \dots, n) \\ D = a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj} \quad (j=1, 2, \dots, n) \end{cases}$$

例. 计算行列式

$$D = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 1 & 3 & -4 \\ 2 & 0 & 1 & -1 \\ 1 & -5 & 3 & -3 \end{vmatrix}$$

$$\begin{aligned} \text{解: } D &\xrightarrow[\substack{C_1-2C_3 \\ C_4+C_3}]{\substack{C_1-2C_3 \\ C_4+C_3}} \begin{vmatrix} 5 & 1 & -1 & 1 \\ -11 & 1 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ -5 & -5 & 3 & 0 \end{vmatrix} = 1 \times (-1)^{3+3} \begin{vmatrix} 5 & 1 & 1 \\ -11 & 1 & -1 \\ -5 & -5 & 0 \end{vmatrix} \xrightarrow{r_2+r_1} \begin{vmatrix} 5 & 1 & 1 \\ -6 & 2 & 0 \\ -5 & -5 & 0 \end{vmatrix} \\ &= 1 \times (-1)^{1+3} \begin{vmatrix} -6 & 2 \\ -5 & -5 \end{vmatrix} = (-6) \times (-5) - 2 \times (-5) = 40 \end{aligned}$$

范德蒙行列式

$$V_n = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix} = \prod_{n \geq i > j \geq 1} (x_i - x_j) = \begin{matrix} (x_2 - x_1)(x_3 - x_1)(x_4 - x_1) \cdots (x_n - x_1) \\ (x_3 - x_2)(x_4 - x_2) \cdots (x_n - x_2) \\ \vdots \\ (x_{n-1} - x_{n-2})(x_n - x_{n-2}) \\ (x_n - x_{n-1}) \end{matrix}$$

推论: 行列式中某一行(列)的各元素与另一行(列)对应元素代数余子式的乘积之和为零

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} D & (i=j) \\ 0 & (i \neq j) \end{cases}$$

$$\text{或 } a_{1i}A_{1j} + a_{2i}A_{2j} + \cdots + a_{ni}A_{nj} = \begin{cases} D & (i=j) \\ 0 & (i \neq j) \end{cases}$$

例. 设 $D = \begin{vmatrix} 3 & -5 & 2 & 1 \\ 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 2 & -4 & -1 & -3 \end{vmatrix}$ 求 $A_{11} + A_{12} + A_{13} + A_{14}$ 及 $M_{11} + M_{21} + M_{31} + M_{41}$

解: 注意到 $A_{11} + A_{12} + A_{13} + A_{14}$ 等于用 $1, 1, 1, 1$ 替代 D 的 r_1 所得行列式

$$\text{即 } A_{11} + A_{12} + A_{13} + A_{14} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 2 & -4 & -1 & -3 \end{vmatrix} \xrightarrow[r_3 - r_1]{r_4 + r_1} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & -5 \\ -2 & 2 & 0 & 2 \\ 1 & -1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -5 & 2 & 1 \\ -2 & 2 & 2 \\ 1 & -1 & 0 \end{vmatrix}$$

$$\xrightarrow{c_2 + c_1} \begin{vmatrix} 1 & 2 & -5 \\ -2 & 0 & 2 \\ 1 & 0 & 0 \end{vmatrix} = -2 \times \begin{vmatrix} -2 & 2 \\ 1 & 0 \end{vmatrix} = 4$$

$$M_{11} + M_{21} + M_{31} + M_{41} = A_{11} - A_{21} + A_{31} - A_{41}$$

$$= \begin{vmatrix} 1 & -5 & 2 & 1 \\ -1 & 1 & 0 & -5 \\ 1 & 3 & 1 & 3 \\ -1 & -4 & -1 & -3 \end{vmatrix} \xrightarrow{r_4 + r_2} \begin{vmatrix} 1 & -5 & 2 & 1 \\ -1 & 1 & 0 & -5 \\ 1 & 3 & 1 & 3 \\ 0 & -1 & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 1 \\ -1 & 0 & -5 \\ 1 & 1 & 3 \end{vmatrix}$$

$$\xrightarrow{r_1 - 2r_3} \begin{vmatrix} -1 & 0 & -5 \\ -1 & 0 & -5 \\ 1 & 1 & 3 \end{vmatrix} = 0$$

小练习

Number 3

1. 求行列式 $\begin{vmatrix} -3 & 0 & 4 \\ 5 & 0 & 3 \\ 2 & -2 & 1 \end{vmatrix}$ 中元素2和-2的代数余子式

$$A_{31} = (-1)^{3+1} \times \begin{vmatrix} 0 & 4 \\ 0 & 3 \end{vmatrix} = 0$$

$$A_{32} = (-1)^{3+2} \times \begin{vmatrix} -3 & 4 \\ 5 & 3 \end{vmatrix} = 29$$

2. 已知四阶行列式D中第三列元素依次为-1, 2, 0, 1, 它们的余子式依次为5, 3, -7, 4, 求D

$$\begin{aligned} D &= a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33} + a_{43}A_{43} \\ &= (-1)^{1+3} \times (-1) \times 5 + (-1)^{2+3} \times 2 \times 3 + (-1)^{3+3} \times 0 \times (-7) + (-1)^{4+3} \times 1 \times 4 \\ &= -15 \end{aligned}$$

3. 证明:

$$(1) \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} = (a-b)^3$$

$$\text{左边} = (-1)^{3+1} \times 1 \times \begin{vmatrix} ab & b^2 \\ a+b & 2b \end{vmatrix} + (-1)^{3+2} \times 1 \times \begin{vmatrix} a^2 & b^2 \\ 2a & 2b \end{vmatrix}$$

$$+ (-1)^{3+3} \times 1 \times \begin{vmatrix} a^2 & ab \\ 2a & a+b \end{vmatrix}$$

$$= ab \cdot 2b - b^2(a+b) - a^2 \cdot 2b + b^2 \cdot 2a + a^2(a+b) - ab \cdot 2a$$

$$= 2ab^2 - ab^2 - b^3 - 2a^2b + 2ab^2 + a^3 + a^2b - 2a^2b$$

$$= a^3 - b^3 + 3ab^2 - 3a^2b$$

$$= (a-b)(a^2 - 2ab + b^2)$$

$$= (a-b)^3$$

= 右边

$$(2) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = (a-b)(a-c)(a-d)(b-c) \cdot (b-d)(c-d)(a+b+c+d)$$

$$\text{左边} \xrightarrow[r_2 - ar_1]{r_4 - ar_3} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b(b-a) & c(c-a) & d(d-a) \\ 0 & b^2(b^2-a^2) & c^2(c^2-a^2) & d^2(d^2-a^2) \end{vmatrix}$$

$$= \begin{vmatrix} b-a & c-a & d-a \\ b(b-a) & c(c-a) & d(d-a) \\ b^2(b^2-a^2) & c^2(c^2-a^2) & d^2(d^2-a^2) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2(b+a) & c^2(c+a) & d^2(d+a) \end{vmatrix}$$

$$\xrightarrow[c_3 - c_1]{c_2 - c_1} (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 0 & 0 \\ b & c-b & d-b \\ b^2(b+a) & c^2(c+a)-b^2(b+a) & d^2(d+a)-b^2(b+a) \end{vmatrix}$$

$$\xrightarrow{\text{按} r_1 \text{ 展开}} (b-a)(c-a)(d-a) \begin{vmatrix} c-b & d-b \\ c^2(c+a)-b^2(b+a) & d^2(d+a)-b^2(b+a) \end{vmatrix}$$

$$\begin{aligned} \therefore c^2(c+a)-b^2(b+a) &= c^2(c+a)-b^2(c+a)+b^2(c+a)-b^2(b+a) \\ &= (c+b)(c-b)(c+a)+b^2(c-b) \\ &= (b^2+c^2+bc+ac+ab)(c-b) \end{aligned}$$

$$\text{同理: } d^2(d+a)-b^2(b+a) = (b^2+d^2+bd+ad+ab)(d-b)$$

$$\begin{aligned} &= (b-a)(c-a)(d-a)(c-b)(d-b) \begin{vmatrix} 1 & 1 \\ b^2+c^2+bc+ac+ab & b^2+d^2+bd+ad+ab \end{vmatrix} \\ &= (b-a)(c-a)(d-a)(c-b)(d-b) [(d^2-c^2)+b(d-c)+a(d-c)] \\ &= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)(a+b+c+d) \end{aligned}$$

4. 设 n 阶行列式 $D = \det(a_{ij})$, 把 D 上下翻转, 或逆时针旋转 90° , 或依副对角线翻转, 依次得

$$D_1 = \begin{vmatrix} a_{n1} & \dots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \dots & a_{1n} \end{vmatrix}, \quad D_2 = \begin{vmatrix} a_{1n} & \dots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \dots & a_{n1} \end{vmatrix}, \quad D_3 = \begin{vmatrix} a_{nn} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{11} \end{vmatrix}$$

证明: $D_1 = D_2 = (-1)^{\frac{n(n-1)}{2}} D$, $D_3 = D$

$$\begin{aligned} D_1 &= \begin{vmatrix} a_{n1} & \dots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \dots & a_{1n} \end{vmatrix} = (-1)^{n-1} \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix} = (-1)^{n-1} \cdot (-1)^{n-2} \dots (-1)^{n-(n-1)} \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix} \\ &= (-1)^{1+2+\dots+n} D = (-1)^{\frac{n(n-1)}{2}} D \end{aligned}$$

$$D_2 = \begin{vmatrix} a_{1n} & \dots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \dots & a_{n1} \end{vmatrix} = (-1)^{\frac{n(n-1)}{2}} \begin{vmatrix} a_{11} & \dots & a_{n1} \\ \vdots & & \vdots \\ a_{1n} & \dots & a_{nn} \end{vmatrix} = (-1)^{\frac{n(n-1)}{2}} D^T = (-1)^{\frac{n(n-1)}{2}} D$$

$$D_3 = \begin{vmatrix} a_{nn} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{11} \end{vmatrix} = (-1)^{\frac{n(n-1)}{2}} D_2 = (-1)^{n(n-1)} D = D.$$

5. 计算下列行列式

$$(1) \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$$

解:

$$\begin{aligned} \text{原式} &= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} + \begin{vmatrix} x & 1 & 1 & 1 \\ 0 & 1-x & 1 & 1 \\ 0 & 1 & 1+y & 1 \\ 0 & 1 & 1 & 1-y \end{vmatrix} \\ &\quad \downarrow r_2-r_1 \quad \downarrow \text{按C}_1 \text{展开} \\ &= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -x & 0 & 0 \\ 0 & 0 & y & 0 \\ 0 & 0 & 0 & -y \end{vmatrix} + x \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1-y \end{vmatrix} \end{aligned}$$

$$= xy^2 + x \{ (1-x)[(1+y)(1-y)-1] + 1 \cdot [1-(1-y)] + 1 \cdot [1-(1+y)] \}$$

$$= xy^2 + x \cdot (1-x)(-y^2) = x^2 y^2$$

$$(2) \begin{vmatrix} 0 & a & b & a \\ a & 0 & a & b \\ b & a & 0 & a \\ a & b & a & 0 \end{vmatrix}$$

解:

$$\begin{aligned} \text{原式} &\xrightarrow{r_1+(r_2+r_3+r_4)} \begin{vmatrix} 2a+b & 2a+b & 2a+b & 2a+b \\ a & 0 & a & b \\ b & a & 0 & a \\ a & b & a & 0 \end{vmatrix} \\ &= (2a+b) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & 0 & a & b \\ b & a & 0 & a \\ a & b & a & 0 \end{vmatrix} \xrightarrow{\substack{r_2-ar_1 \\ r_3-br_1 \\ r_4-ar_1}} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -a & 0 & b-a \\ 0 & -a & -b-a & 0 \\ 0 & b-a & 0 & -a \end{vmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -a & -b-a & 0 \\ 0 & -a & 0 & b-a \\ 0 & b-a & 0 & -a \end{vmatrix} \\ &\xrightarrow{r_4+r_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -a & -b-a & 0 \\ 0 & -a & 0 & b-a \\ 0 & 0 & -b & -b \end{vmatrix} \xrightarrow{r_3+r_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -a & -b-a & 0 \\ 0 & 0 & -b & -b \\ 0 & 0 & -b & -b \end{vmatrix} \xrightarrow{r_4-r_3} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -a & -b-a & 0 \\ 0 & 0 & -b & -b \\ 0 & 0 & 0 & 0 \end{vmatrix} \end{aligned}$$

$$\xrightarrow{r_2 \leftrightarrow r_3} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -a & 0 & b-a \end{vmatrix} \xrightarrow{r_3 \leftrightarrow r_4} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -a & 0 & b-a \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

$$\xrightarrow{r_4+ar_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & a & b-a \end{vmatrix} \xrightarrow{r_4-ar_3} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & b-2a \end{vmatrix}$$

$$= (2a+b) b^2 (b-2a) = (b^2-4a^2) b^2 = b^4-4a^2 b^2$$

$$(3) \begin{vmatrix} x & y & 0 & \cdots & 0 & 0 \\ 0 & x & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & y \\ y & 0 & 0 & \cdots & 0 & x \end{vmatrix}$$

解:

$$\begin{aligned} \text{原式} &= (-1)^{n+1} \cdot x \cdot \begin{vmatrix} x & y & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & x & y \\ 0 & 0 & \cdots & 0 & x \end{vmatrix} + (-1)^{n+1} \cdot y \cdot \begin{vmatrix} y & 0 & \cdots & 0 & 0 \\ x & y & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & x & y \end{vmatrix} \\ &= x \cdot x^{n-1} + (-1)^{n+1} \cdot y \cdot y^{n-1} \\ &= x^n + (-1)^{n+1} y^n \end{aligned}$$

$$(4) \begin{vmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & a_1+x \end{vmatrix}$$

解: 原式 $\frac{C_{i+1} + x C_i}{(i=n, \dots, 2)}$

$$\begin{vmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -1 \\ a_n & a_{n-1} & a_{n-2} + a_{n-3}x + \cdots & & a_2 + a_1x + x^2 & a_1+x \\ + a_{n-1}x & + a_{n-2}x & a_{n-4}x^2 + \cdots & & & \\ + \cdots & + \cdots & a_1x^{n-3} + x^{n-2} & & & \\ + a_1x^{n-1} & + a_1x^{n-2} & & & & \\ + x^n & + x^{n-1} & & & & \end{vmatrix}$$

$$\begin{aligned} &= (-1)^{n+1} (a_n + a_{n-1}x + \cdots + a_1x^{n-1} + x^n) \cdot (-1)^{n-1} \\ &= x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n \end{aligned}$$

$$(5) \begin{vmatrix} -a_1 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_n & a_n \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{vmatrix}$$

解:

$$\text{原式} \frac{C_{i+1} + C_i}{(i=1, 2, \dots, n-1)} \begin{vmatrix} -a_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & -a_3 & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_n & 0 \\ 1 & 2 & 3 & \cdots & 1 & n+1 \end{vmatrix}$$

$$\begin{aligned} &= (-a_1)(-a_2) \cdots (-a_n)(n+1) \\ &= (-1)^n (n+1) a_1 a_2 \cdots a_n \end{aligned}$$

$$(6) \begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ 2 & 3 & 4 & \cdots & 1 \\ 3 & 4 & 5 & \cdots & 2 \\ \vdots & \vdots & \vdots & & \vdots \\ n & 1 & 2 & \cdots & n-1 \end{vmatrix}$$

解: 原式 $\frac{C_1 + C_2 + \cdots + C_n}{(i=1, 2, \dots, n-1)}$

$$\begin{aligned} &= \frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ 1 & 3 & 4 & \cdots & 1 \\ 1 & 4 & 5 & \cdots & 2 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 2 & \cdots & n-1 \end{vmatrix} \frac{r_i - r_{i-1}}{(i=2, \dots, n)} \frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ 0 & 1 & 1 & \cdots & 1-n \\ 0 & 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 1-n & 1 & \cdots & 1 \end{vmatrix} \\ &= \frac{n(n+1)}{2} \cdot 1 \cdot (-1)^{n+1} \begin{vmatrix} 1 & 1 & \cdots & 1-n \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1-n & 1 & \cdots & 1 \end{vmatrix} \\ &= \frac{n(n+1)}{2} \cdot \frac{C_1 + \cdots + C_{n-1}}{n(n+1)} \begin{vmatrix} -1 & 1 & \cdots & 1-n \\ -1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ -1 & 1 & \cdots & 1 \end{vmatrix} \frac{C_i + C_1}{(i=2, \dots, n-1)} \frac{n(n+1)}{2} \begin{vmatrix} -1 & 0 & \cdots & -n \\ -1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ -1 & 0 & \cdots & 0 \end{vmatrix} \\ &\text{按 } r_{n-1} \text{ 展开} \quad \frac{n(n+1)}{2} \cdot (-1)^{n-1} \cdot (-1)^{n-1+1} \cdot (-1)^{\frac{(n-2)(n-3)}{2}} \cdot (-n)^{n-2} \\ &= (-1)^{\frac{n^2-n+4}{2}} \frac{n^2+n-1}{2} = (-1)^{\frac{n(n-1)}{2}} \frac{n^n + n^{n-1}}{2} \end{aligned}$$

6. 计算下列行列式 (D_n 为 n 阶行列式)

$$(1) D_n = \begin{vmatrix} a & 0 & 0 & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & 0 & 0 \\ 0 & 0 & a & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a & 0 \\ 1 & 0 & 0 & \cdots & 0 & a \end{vmatrix}$$

解: 原式 按 r_n 展开

$$1 \cdot (-1)^{n+1} \begin{vmatrix} 0 & 0 & \cdots & 0 & 1 \\ a & 0 & \cdots & 0 & 0 \\ 0 & a & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a & 0 \end{vmatrix} + a \cdot (-1)^{n+n} \begin{vmatrix} a & 0 & 0 & \cdots & 0 \\ 0 & a & 0 & \cdots & 0 \\ 0 & 0 & a & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a \end{vmatrix}$$

$$= (-1)^{n+1} \cdot 1 \cdot (-1)^{1+n-1} \begin{vmatrix} a & 0 & \cdots & 0 \\ 0 & a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a \end{vmatrix} + a \cdot a^{n-1} = -a^{n-2} + a^n$$

$$(2) D_{n+1} = \begin{vmatrix} a^n & (a-1)^n & \cdots & (a-n)^n \\ a^{n-1} & (a-1)^{n-1} & \cdots & (a-n)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a & a-1 & \cdots & a-n \\ 1 & 1 & \cdots & 1 \end{vmatrix}$$

解: 原式 $= (-1)^n \cdot (-1)^{n-1} \cdots (-1)^1 \begin{vmatrix} 1 & 1 & \cdots & 1 \\ a & a-1 & \cdots & a-n \\ \vdots & \vdots & \ddots & \vdots \\ a^{n-1} & (a-1)^{n-1} & \cdots & (a-n)^{n-1} \\ a^n & (a-1)^n & \cdots & (a-n)^n \end{vmatrix} = (-1)^{\frac{n(n+1)}{2}} \prod_{0 \leq i < j \leq n} [(a-j) - (a-i)] = (-1)^{\frac{n(n+1)}{2}} \prod_{0 \leq i < j \leq n} (i-j)$

$$(3) D_{2n} = \begin{vmatrix} a_n & \cdots & b_n \\ & \ddots & \vdots \\ & a_1 & b_1 \\ & c_1 & d_1 \\ & \vdots & \vdots \\ c_n & \cdots & d_n \end{vmatrix}$$

解: 原式 按 r_1 展开

$$a_n (-1)^{1+1} \begin{vmatrix} a_{n-1} & \cdots & b_{n-1} & 0 \\ & \ddots & \vdots & \vdots \\ & a_1 & b_1 & \vdots \\ & c_1 & d_1 & \vdots \\ & \vdots & \vdots & \vdots \\ c_{n-1} & \cdots & d_{n-1} & d_n \end{vmatrix} + b_n (-1)^{1+2n} \begin{vmatrix} 0 & a_{n-1} & \cdots & b_{n-1} \\ & \ddots & \vdots & \vdots \\ & a_1 & b_1 & \vdots \\ & c_1 & d_1 & \vdots \\ & \vdots & \vdots & \vdots \\ c_n & \cdots & d_{n-1} & 0 \end{vmatrix}$$

按最后一行展开

$$a_n \cdot d_n D_{2(n-1)} - b_n \cdot c_n \cdot D_{2(n-1)} \Rightarrow D_{2n} = (a_n d_n - b_n c_n) D_{2n-2}$$

$$D_{2n} = \prod_{i=2}^n (a_i d_i - b_i c_i) D_2 \quad \text{而 } D_2 = \begin{vmatrix} a_1 & b_1 \\ c_1 & d_1 \end{vmatrix} = a_1 d_1 - b_1 c_1$$

$$\therefore D_{2n} = \prod_{i=1}^n (a_i d_i - b_i c_i)$$

(4) $D_n = \det(a_{ij})$, 其中 $a_{ij} = |i-j|$

解: $D_n = \begin{vmatrix} 0 & 1 & 2 & \dots & n-1 \\ 1 & 0 & 1 & \dots & n-2 \\ 2 & 1 & 0 & \dots & n-3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n-1 & n-2 & n-3 & \dots & 0 \end{vmatrix} \xrightarrow[r_i - r_{i-1}]{(i=n, \dots, 2)} \begin{vmatrix} 0 & 1 & 2 & \dots & n-1 \\ 1 & -1 & -1 & \dots & -1 \\ 1 & 1 & -1 & \dots & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & -1 \end{vmatrix} \xrightarrow[C_i + C_n]{(i=1, \dots, n-1)} \begin{vmatrix} n-1 & n & n+1 & \dots & n-1 \\ 0 & -2 & -2 & \dots & -1 \\ 0 & 0 & -2 & \dots & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -1 \end{vmatrix} = (n-1) \cdot 2^{n-2} \cdot (-1)^n$

(5) $D_n = \begin{vmatrix} 1+a_1 & 1 & \dots & 1 \\ 1 & 1+a_2 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1+a_n \end{vmatrix}$, 其中 $a_1 a_2 \dots a_n \neq 0$

解: $D_n \xrightarrow[C_i - C_{i+1}]{(i=1, \dots, n-1)} \begin{vmatrix} a_1 & 0 & 0 & \dots & 0 & 0 & 1 \\ -a_2 & a_2 & 0 & \dots & 0 & 0 & 1 \\ 0 & -a_3 & a_3 & \dots & 0 & 0 & 1 \\ 0 & 0 & -a_4 & \dots & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -a_{n-1} & a_{n-1} & 1 \\ 0 & 0 & 0 & \dots & 0 & -a_n & 1+a_n \end{vmatrix}$

按最后一列从下往上展开

$$\begin{aligned} & (1+a_n) \cdot (-1)^{n+n} (a_1 a_2 \dots a_{n-1}) + (-1)^{n-1+n} [a_1 a_2 \dots a_{n-2} \cdot (-a_n)] \\ & + (-1)^{n-2+n} [a_1 a_2 \dots a_{n-3} \cdot (-a_{n-1}) \cdot (-a_n)] \dots \\ & = a_1 a_2 \dots a_{n-1} \dot{V} + \underline{a_1 a_2 \dots a_n} + a_1 a_2 \dots a_{n-2} \dot{V} a_n + a_1 a_2 \dots a_{n-3} \dot{V} a_{n-1} a_n \\ & \quad + \dots \dot{V} a_2 a_3 \dots a_n \\ & = (a_1 a_2 \dots a_n) \left(1 + \sum_{i=1}^n \frac{1}{a_i} \right) \end{aligned}$$

克莱姆法则

二元线性方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$
, 当系数行列式 $D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} \neq 0$ 时,

方程组的解可记为
$$x_1 = \frac{D_1}{D} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{D_2}{D} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

n元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \dots \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{cases}$$

当 $b_1, b_2, \dots, b_n$ 不全为零时, 称为非齐次线性方程组; 当 $b_1, b_2, \dots, b_n$ 全为零时, 称为齐次线性方程组

$D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$ 称为该方程组的系数行列式

若线性方程组的系数行列式 $D \neq 0$, 则此方程组有唯一解

$$x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad \dots, \quad x_n = \frac{D_n}{D}$$

其中 $D_j (j=1, 2, \dots, n)$ 是将 D 中第 j 列元素用右端项 $b_1, \dots, b_n$ 代替后所得行列式:

$$D_j = \begin{vmatrix} a_{11} & \dots & a_{1,j-1} & b_1 & a_{1,j+1} & \dots & a_{1n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \dots & a_{n,j-1} & b_n & a_{n,j+1} & \dots & a_{nn} \end{vmatrix}$$

例. 用克莱姆法则求解方程组

$$\begin{cases} 2x_1 + x_2 - 5x_3 + x_4 = 8 \\ x_1 - 3x_2 \quad \quad - 6x_4 = 9 \\ 2x_2 - x_3 + 2x_4 = -5 \\ x_1 + 4x_2 - 7x_3 + 6x_4 = 0 \end{cases}$$

解: $D = \begin{vmatrix} 2 & 1 & -5 & 1 \\ 1 & -3 & 0 & -6 \\ 0 & 2 & -1 & 2 \\ 1 & 4 & -7 & 6 \end{vmatrix} \xrightarrow[r_4 - r_2]{r_1 - 2r_2} \begin{vmatrix} 0 & 7 & -5 & 13 \\ 1 & -3 & 0 & -6 \\ 0 & 2 & -1 & 2 \\ 0 & 7 & -7 & 12 \end{vmatrix} \xrightarrow{\text{按 } C_1 \text{ 展开}} 1 \times (-1)^{2+1} \times \begin{vmatrix} 7 & -5 & 13 \\ 2 & -1 & 2 \\ 7 & -7 & 12 \end{vmatrix} = - \begin{vmatrix} 7 & -5 & 13 \\ 2 & -1 & 2 \\ 7 & -7 & 12 \end{vmatrix}$

$$\xrightarrow[C_3 + 2C_2]{C_1 + 2C_2} - \begin{vmatrix} -3 & -5 & 3 \\ 0 & -1 & 0 \\ -7 & -7 & -2 \end{vmatrix} \xrightarrow{\text{按 } r_2 \text{ 展开}} -(-1) \times (-1)^{2+2} \times \begin{vmatrix} -3 & 3 \\ -7 & -2 \end{vmatrix} = 27$$

$D_1 = \begin{vmatrix} 8 & 1 & -5 & 1 \\ 9 & -3 & 0 & -6 \\ -5 & 2 & -1 & 2 \\ 0 & 4 & -7 & 6 \end{vmatrix} = 81, \quad D_2 = \begin{vmatrix} 2 & 8 & -5 & 1 \\ 1 & 9 & 0 & -6 \\ 0 & -5 & -1 & 2 \\ 1 & 0 & -7 & 6 \end{vmatrix} = -108, \quad D_3 = \begin{vmatrix} 2 & 1 & 8 & 1 \\ 1 & -3 & 9 & -6 \\ 0 & 2 & -5 & 2 \\ 1 & 4 & 0 & 6 \end{vmatrix} = -27, \quad D_4 = \begin{vmatrix} 2 & 1 & -5 & 8 \\ 1 & -3 & 0 & 9 \\ 0 & 2 & -1 & -5 \\ 1 & 4 & -7 & 0 \end{vmatrix} = 27$

$$\therefore x_1 = \frac{D_1}{D} = 3, \quad x_2 = \frac{D_2}{D} = -4, \quad x_3 = \frac{D_3}{D} = -1, \quad x_4 = \frac{D_4}{D} = 1$$

若线性方程组的系数行列式 $D \neq 0$, 则此方程组一定有解, 且解是唯一的.

[逆否定理] 若线性方程组无解或有两个不同的解, 则其系数行列式必为 0

对齐次线性方程组, 易见 $x_1 = x_2 = \dots = x_n = 0$ 显然是该方程组的解, 称之为零解.

若齐次线性方程组的系数行列式 $D \neq 0$, 则此方程组一定仅有零解, 而无非零解

[逆否定理] 若齐次线性方程组有非零解, 则其系数行列式必为 0

系数行列式为零 $\Leftrightarrow$ 齐次线性方程组有非零解

例. 问当 λ 为何值时, 齐次线性方程组
$$\begin{cases} (1-\lambda)x_1 - 2x_2 + 4x_3 = 0 \\ 2x_1 + (3-\lambda)x_2 + x_3 = 0 \\ x_1 + x_2 + (1-\lambda)x_3 = 0 \end{cases}$$
 有非零解?

解: 系数行列式
$$D = \begin{vmatrix} 1-\lambda & -2 & 4 \\ 2 & 3-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} \xrightarrow{C_2-C_1} \begin{vmatrix} 1-\lambda & \lambda-3 & 4 \\ 2 & 1-\lambda & 1 \\ 1 & 0 & 1-\lambda \end{vmatrix} = 1 \times (-1)^{3+1} \begin{vmatrix} \lambda-3 & 4 \\ 1-\lambda & 1 \end{vmatrix} + (1-\lambda) \times (-1)^{3+3} \begin{vmatrix} 1-\lambda & \lambda-3 \\ 2 & 1-\lambda \end{vmatrix}$$

$$= (\lambda-3) - 4(1-\lambda) + (1-\lambda)[(1-\lambda)^2 - 2(\lambda-3)] = (1-\lambda)^3 - 2(1-\lambda)(\lambda-3) + (\lambda-3) - 4(1-\lambda)$$

$$= (1-\lambda)^3 + (2\lambda-1)(\lambda-3) - 4(1-\lambda) = -\lambda^3 + 5\lambda^2 - 6\lambda = \lambda(-\lambda^2 + 5\lambda - 6) = -\lambda(\lambda-2)(\lambda-3)$$

令 $D=0$, 则 $\lambda=0, \lambda=2$ 或 $\lambda=3$ 时, 有非零解

例. 证明 过 $(n+1)$ 个不同点的 n 次多项式存在且唯一 (克莱姆法则 + 范德蒙行列式)

证明:

设过 $(n+1)$ 个不同点 $(x_i, y_i) (i=0, 1, 2, \dots, n)$ 的多项式为 $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$,

$$\text{则 } \begin{cases} a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n = y_0 \\ a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n = y_1 \\ \dots \dots \dots \\ a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n = y_n \end{cases}$$

其系数行列式为范德蒙行列式

$$D_n = \begin{vmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{vmatrix} \xrightarrow{D_n^T = \dots} = \prod_{0 \leq i < j \leq n} (x_j - x_i) \neq 0$$

由克莱姆法则, 线性方程组存在唯一解,

即所求多项式存在且唯一